

# HR Interview & Meeting Analyzer

Enterprise-Grade AI Dashboard for Audio Intelligence

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# 1 Introduction

Recruitment interviews, corporate meetings, and professional discussions generate large volumes of unstructured audio data. Manual evaluation is time-consuming, subjective, and difficult to scale.

This project presents an **AI-powered HR Interview & Meeting Analyzer** that automatically:

- Transcribes spoken audio into text.
- Segments conversations with speaker identification.
- Performs sentiment and emotion analysis.
- Extracts keywords and enables audio-linked navigation.
- Generates visual analytics for recruiter insights.
- Produces AI-generated summaries and hiring recommendations.

The system is implemented using **Python, Whisper ASR, NLTK, Streamlit, and Plotly**, delivering a professional, enterprise-style dashboard.

## 2 System Architecture

### 2.1 Pipeline Overview

1. **Audio Upload:** User uploads an interview or meeting audio file.
2. **Speech-to-Text (ASR):** Whisper model converts audio into timestamped transcripts.
3. **Segmentation:** Audio is divided into logical speech segments.
4. **Speaker Identification:** Each segment is labeled as Interviewer, Candidate, or Narrator.
5. **Sentiment & Emotion Analysis:** VADER sentiment and heuristic emotion classification.
6. **Keyword Extraction:** TF-IDF-based keyword detection.
7. **Visualization & Decision Logic:** Interactive dashboards and recruiter recommendations.

### 2.2 Technology Stack

- **Frontend:** Streamlit
- **Speech Recognition:** OpenAI Whisper
- **NLP:** NLTK, TF-IDF
- **Visualization:** Plotly, Matplotlib
- **Audio Processing:** Pydub

## 3 Repository and README Integration

### 3.1 GitHub Repository

This project is version-controlled and documented in the GitHub repository:

**Repository URL:**

<https://github.com/springboardmentor13579x-proj/Automated-Podcast-Transcription-and-T>

### 3.2 README Structure

The README.md file contains:

- Project overview and objectives
- Directory structure
- Environment setup instructions
- Execution commands for backend and UI
- Example outputs and screenshots

### 3.3 Execution Commands

**Run backend pipeline:**

```
python -m src.main
```

**Launch UI dashboard:**

```
streamlit run src/ui_app.py
```

## 4 User Interface Overview

The dashboard is organized into modular tabs:

- Upload
- Transcript
- Segments
- Visual Analytics
- Keywords
- Sentiment Metrics
- AI Summary

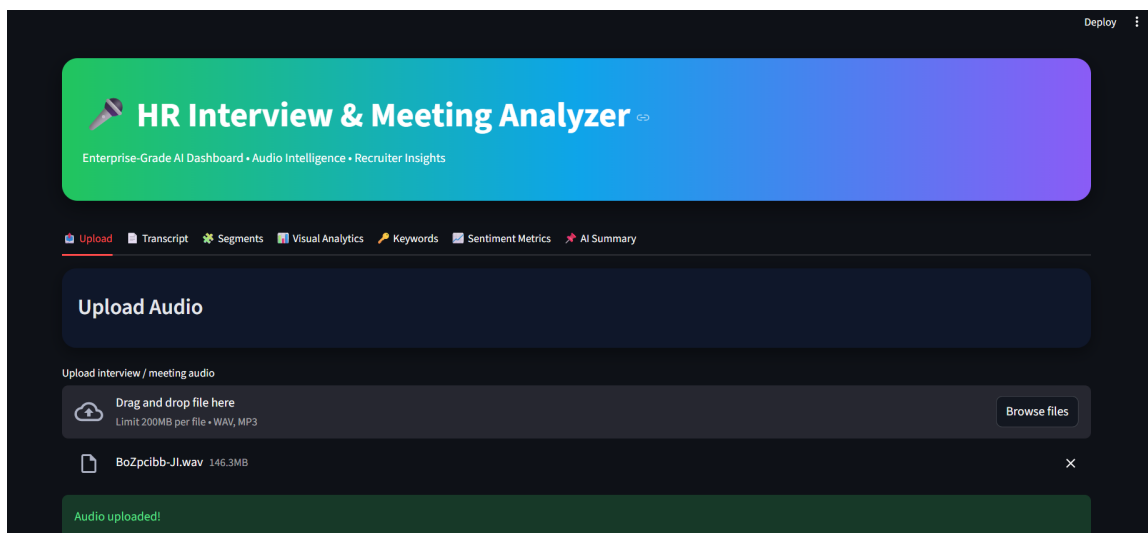


Figure 1: HR Interview & Meeting Analyzer Dashboard

## 5 Transcript and Segmentation

### 5.1 Full Transcript

The system displays the complete transcript of the uploaded audio, allowing users to review the entire conversation.

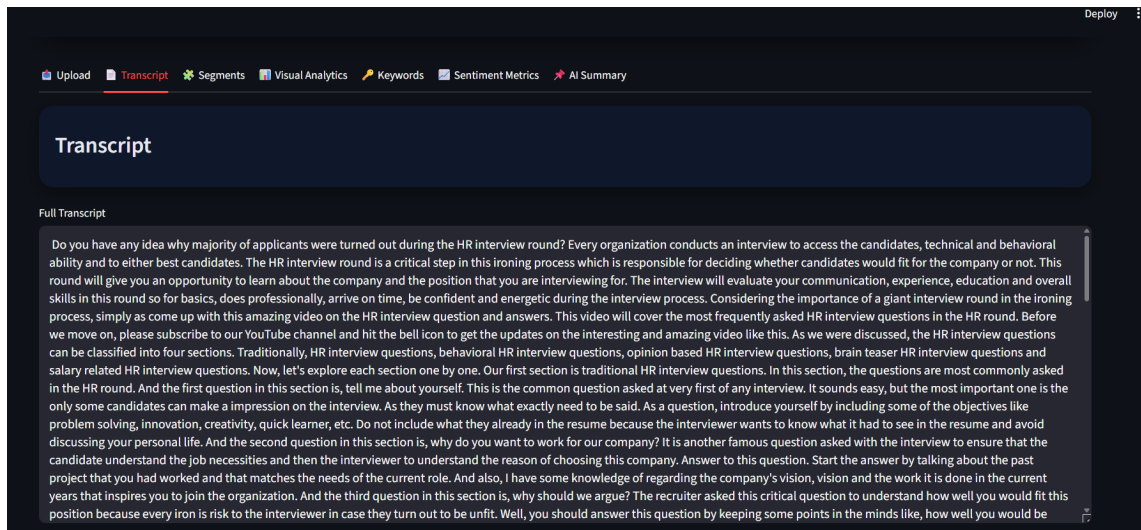


Figure 2: Automatically Generated Full Transcript

### 5.2 Conversation Segments

Each segment includes:

- Speaker role (Interviewer / Candidate / Narrator)
- Timestamp range
- Spoken text
- Embedded audio clip for instant playback

The system enforces conversation structure: once the interview begins, narrator segments are excluded except for introduction and conclusion.

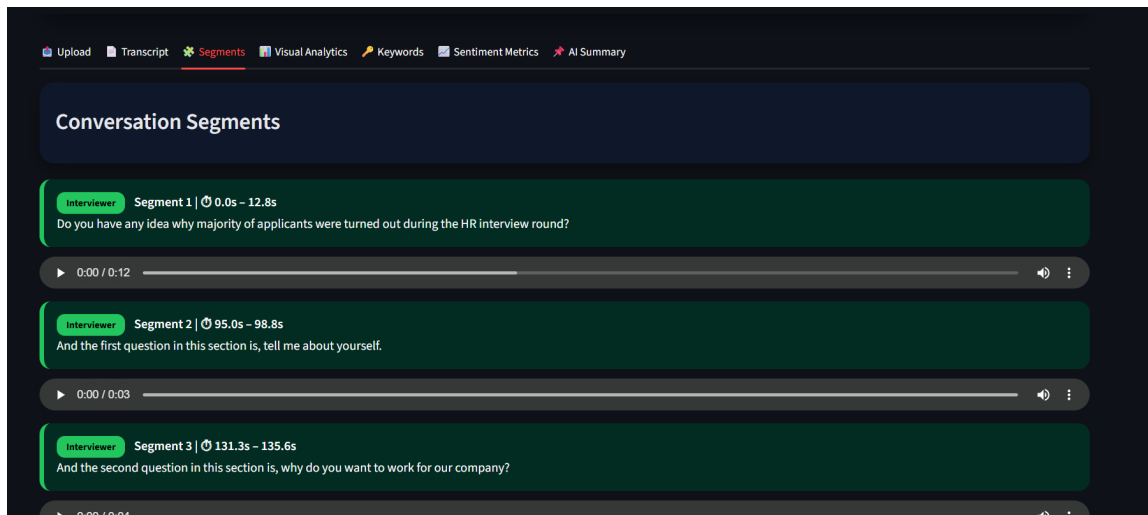


Figure 3: Color-Coded Segments with Audio Playback

## 6 Visual Analytics

This module provides six professional-grade analytics.

### 6.1 1. Sentiment Timeline

Tracks sentiment variation across segments and speakers.

**Interpretation:** Each point represents a spoken segment. Positive values indicate confidence and enthusiasm, while negative values reflect stress or hesitation.

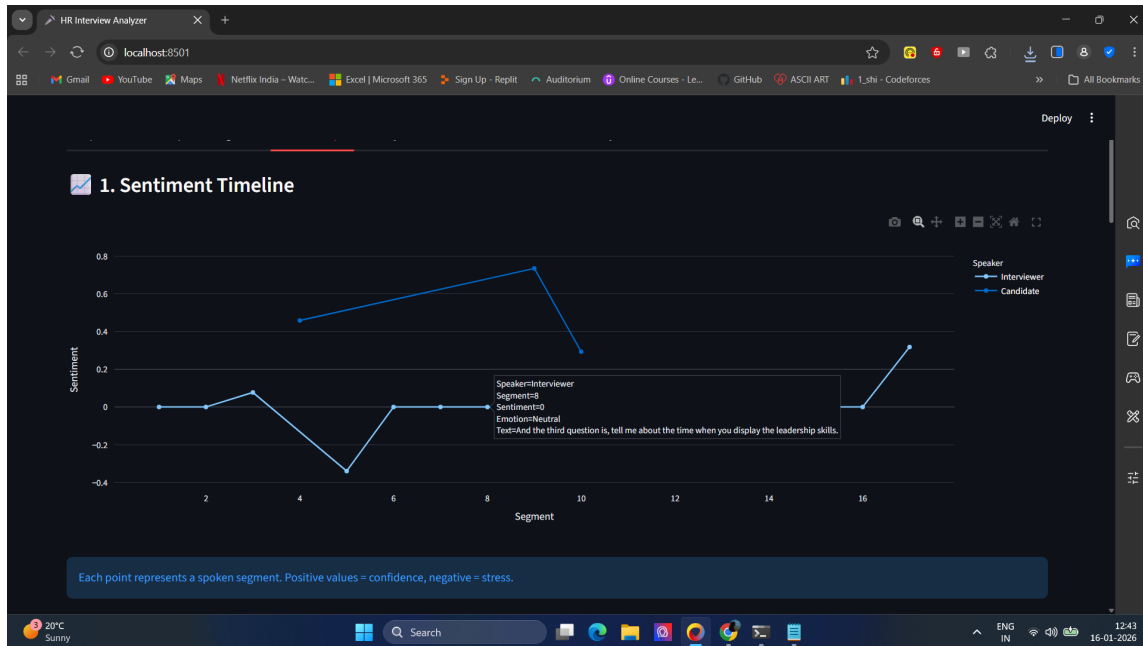


Figure 4: Sentiment Trend Across Interview Segments

### 6.2 2. Segment Polarity Heatmap

Displays sentiment intensity using color coding.

**Interpretation:**

- **Blue:** Confident / positive responses
- **Red:** Negative or hesitant answers
- Light shades indicate neutrality

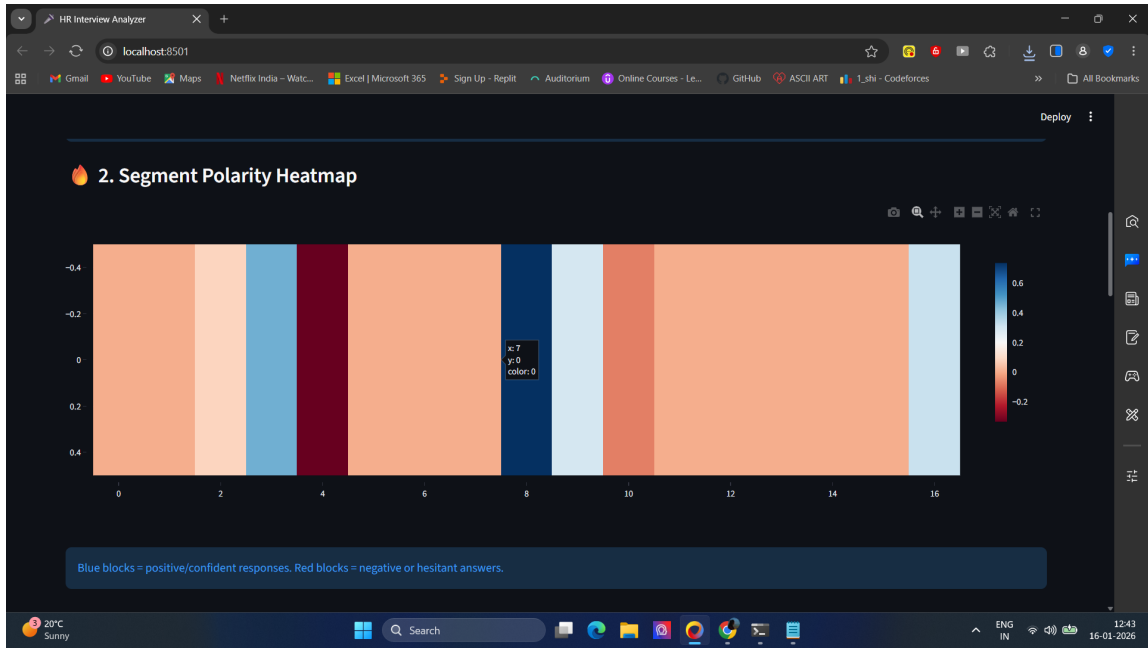


Figure 5: Segment Polarity Heatmap

### 6.3 3. Speaker Distribution

Shows participation level of each speaker.

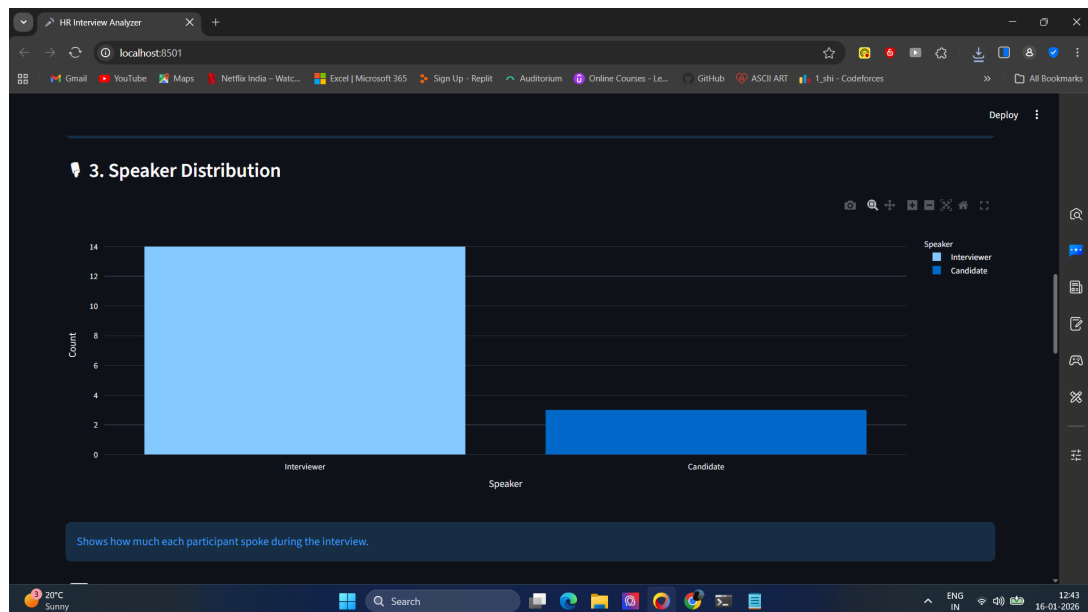


Figure 6: Speaker Contribution Analysis

## 6.4 4. Average Sentiment by Speaker

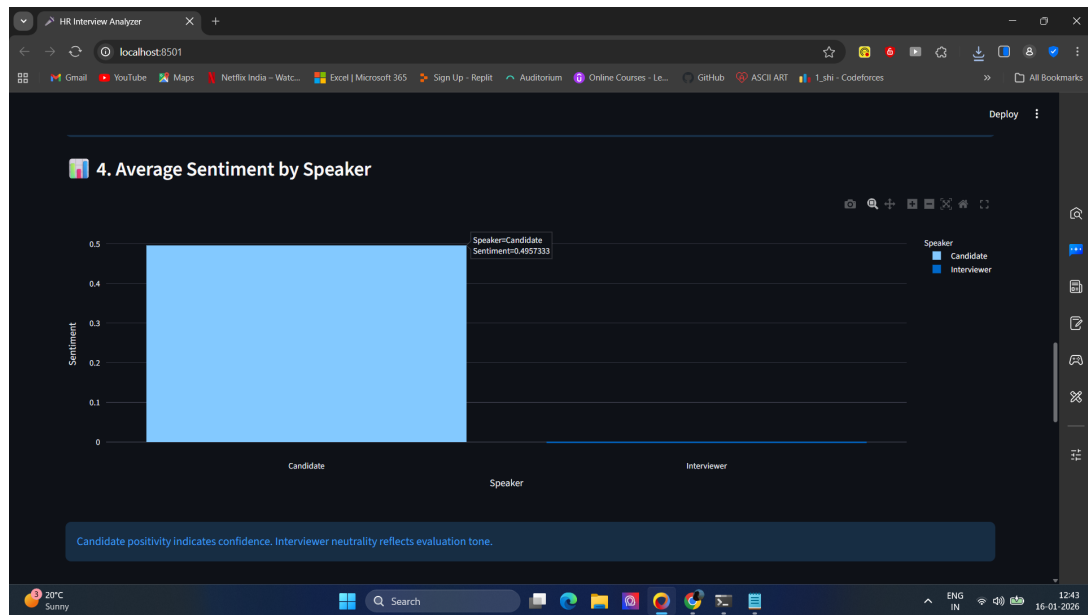


Figure 7: Average Sentiment by Role

## 6.5 5. Emotion Classification

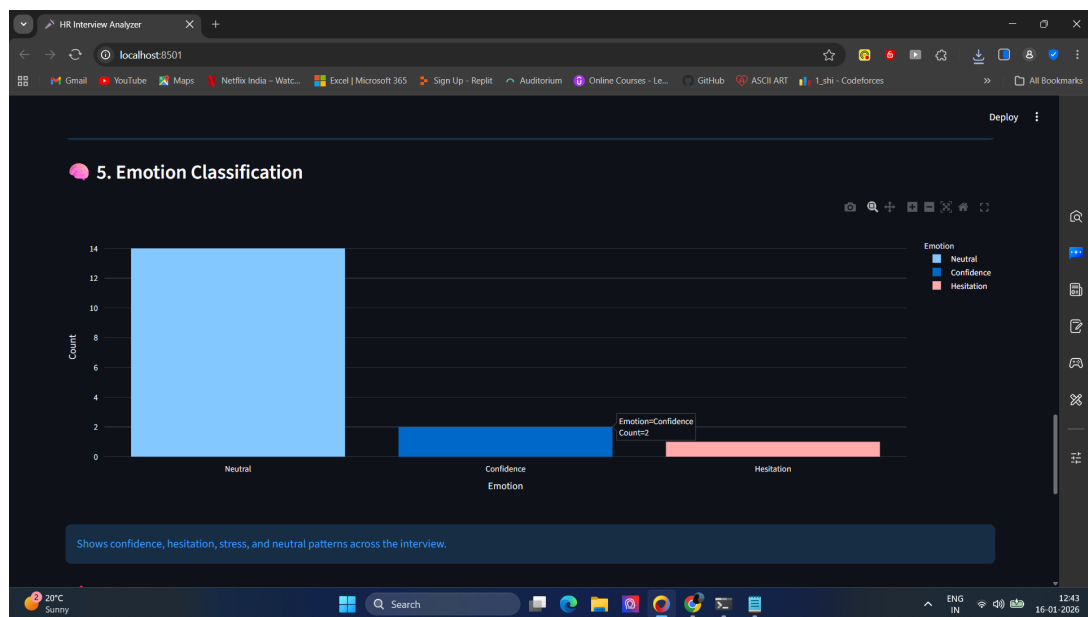


Figure 8: Emotion Distribution

## 6.6 6. Candidate Engagement Radar

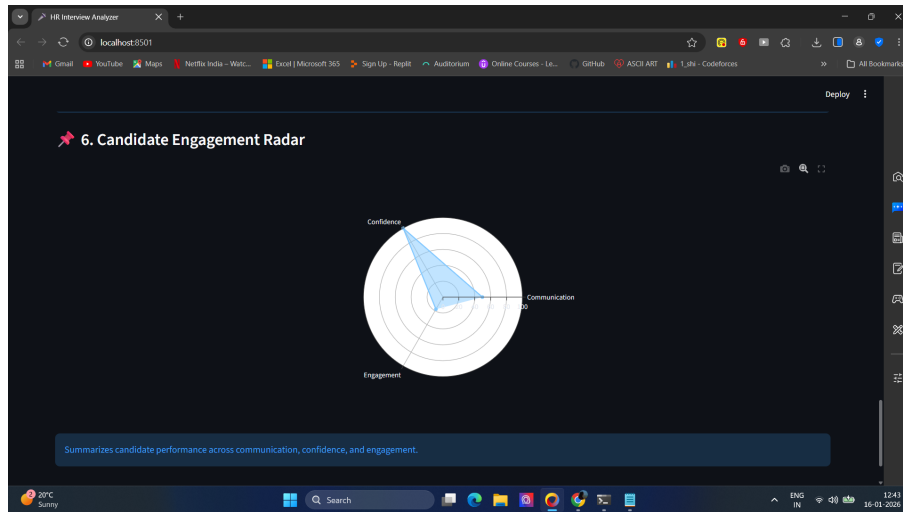


Figure 9: Candidate Engagement Radar

## 7 Keyword Intelligence

### 7.1 Keyword Search with Audio Navigation

Selecting a keyword:

- Filters relevant segments
- Displays timestamps
- Enables direct audio playback

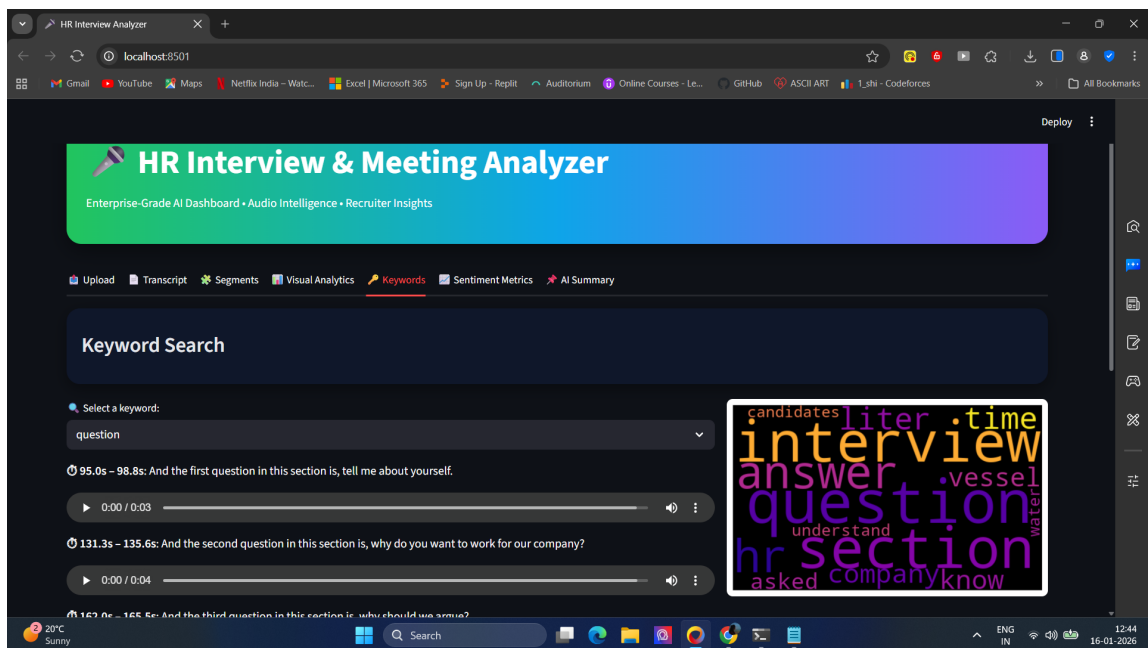


Figure 10: Keyword-Based Segment Navigation

### 7.2 Keyword Cloud



Figure 11: Keyword Cloud Visualization

## 8 Sentiment Metrics and Recruiter Decision

### 8.1 Overall Metrics

- Average Sentiment Score
- Positive Percentage
- Negative Percentage
- Overall Emotional Tone

### 8.2 Recruiter Decision Logic

- **Hire:** Strong positive indicators
- **Hold:** Moderate potential, needs further evaluation
- **Reject:** Predominantly negative or low engagement

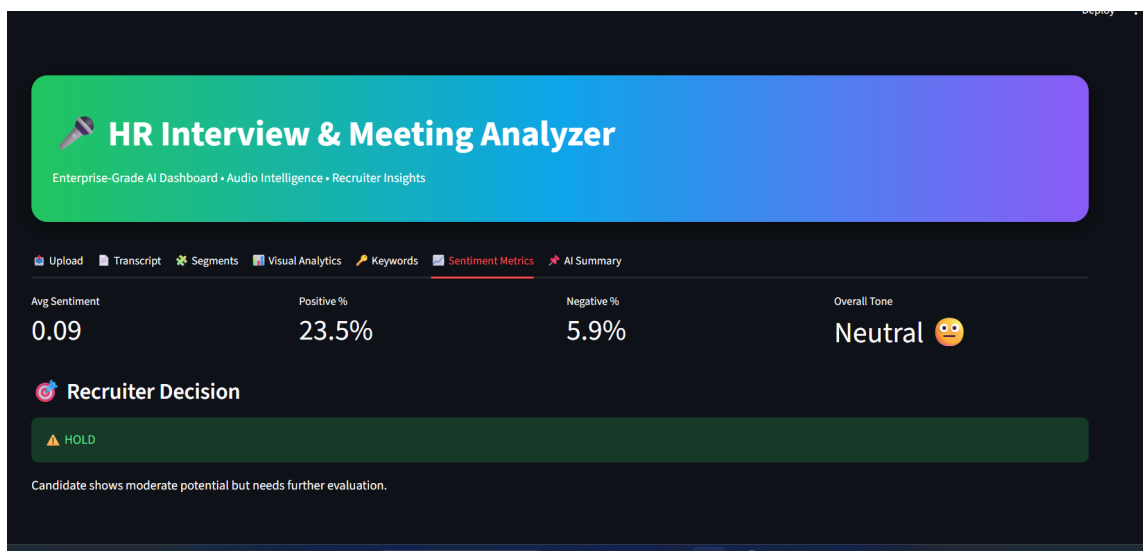


Figure 12: Automated Recruiter Decision and Justification

## 9 AI-Generated Summary

The system produces a concise AI-generated summary capturing:

- Key discussion points
- Candidate strengths and weaknesses
- Overall session conclusion

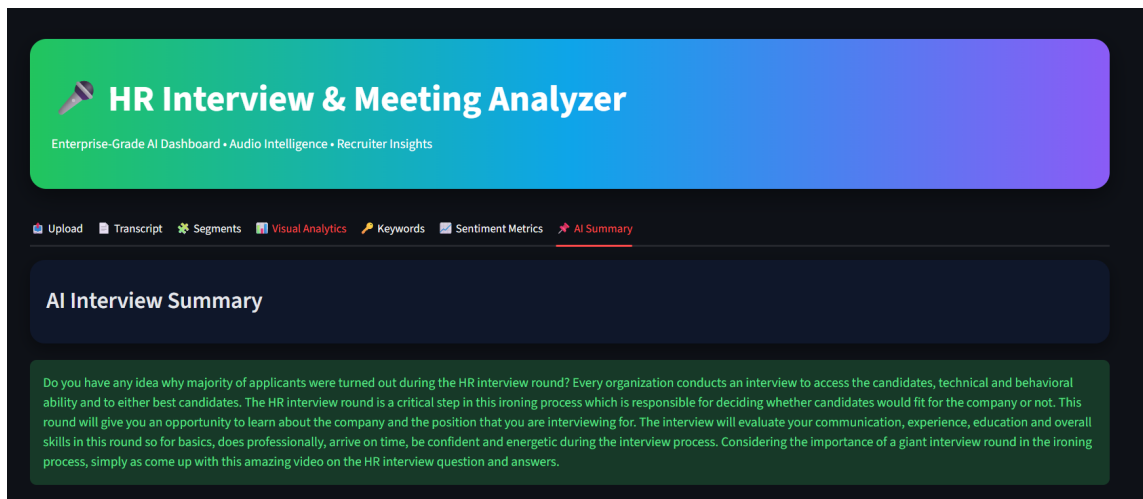


Figure 13: AI Interview Summary

## 10 System Outputs

The system generates structured outputs at each stage of the pipeline. All outputs are documented in the repository and referenced in the project README.

### 10.1 Generated Files

- `transcripts/asr/*.txt` – Raw ASR transcripts generated using Whisper.
- `segments/*.txt` – Speaker-labeled and timestamped conversation segments.
- `docs/keywords.txt` – Extracted keywords from the transcript.
- `docs/asr_evaluation.csv` – Word Error Rate (WER) and transcription accuracy metrics.
- `docs/asr_evaluation_table.txt` – Tabulated ASR performance summary.
- `results/Documentation.Report.md` – Markdown version of the project documentation.

### 10.2 Example Audio: HR Interview Sample

The audio file `BoZpcibb-JI.txt` was processed through the complete pipeline.

#### 10.2.1 Extracted Keywords

interview, round, candidates, company, hr, ability, access, applicants

#### 10.2.2 ASR Accuracy and Evaluation

| File            | WER    | CER    | Accuracy (%) |
|-----------------|--------|--------|--------------|
| BoZpcibb-JI.txt | 0.2099 | 0.1278 | 79.01        |

**Interpretation:** The Word Error Rate (WER) of 0.2099 and Character Error Rate (CER) of 0.1278 demonstrate reliable transcription quality. An overall accuracy of **79.01%** confirms that the system produces sufficiently precise transcripts for segmentation, keyword extraction, sentiment analysis, and recruiter decision support.

### 10.3 Recruiter Decision Output

**Recruiter Decision: HOLD**

**Justification:** The interview demonstrates balanced communication with moderate positivity and low negative sentiment. While keyword relevance and transcription accuracy confirm meaningful responses, emotional and engagement indicators suggest further evaluation is required before final selection.

## **11 Use Cases**

### **11.1 HR Interviews**

- Automated candidate evaluation
- Audio-linked response review
- Bias-reduced recruiter insights

### **11.2 Corporate Meetings**

- Speaker participation analysis
- Sentiment-driven decision insights

### **11.3 Podcasts and Lectures**

- Topic navigation
- Accessibility

## 12 Conclusion

- This project demonstrates how Artificial Intelligence can transform unstructured audio from HR interviews, meetings, and professional discussions into structured, analyzable, and decision-ready information.
- The system integrates a complete **backend pipeline**—including Whisper-based transcription, segmentation, keyword extraction, sentiment and emotion analysis, and evaluation metrics—with an interactive **frontend dashboard**.
- The Streamlit interface enables audio-linked navigation, professional visual analytics, keyword intelligence, and automated recruiter decision support.
- Experimental results on real interview data (e.g., `BoZpcibb-JI.txt`) show reliable transcription performance (WER = 0.2099, Accuracy = 79.01%), meaningful keyword extraction, and emotionally informed insights.
- The automated hiring recommendation (*HOLD*) illustrates the system’s capability to support structured, fair, and data-driven recruitment decisions.
- Overall, the **HR Interview & Meeting Analyzer** bridges the gap between raw audio and actionable intelligence, providing a scalable, enterprise-ready AI solution for modern communication analytics.